

Pharmacokinetic and Pharmacodynamic Modeling of Tezepelumab to Guide Phase 3 Dose Selection for Patients With Severe Asthma

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Abstract

Tezepelumab is a human monoclonal antibody that blocks thymic stromal lymphopoietin, an epithelial cytokine involved in asthma pathogenesis. In the phase 2b PATHWAY study (ClinicalTrials.gov identifier: NCT02054130), tezepelumab significantly reduced exacerbations in adults with severe, uncontrolled asthma. We used pharmacokinetic (PK) and pharmacodynamic (PD) modeling to guide tezepelumab dose selection for phase 3 trials in patients with severe asthma. PK data from 7 clinical studies were used to develop a population PK model. Population PK-PD models were developed to characterize the relationship between tezepelumab PK and asthma exacerbation rate (AER) and fractional exhaled nitric oxide (FeNO) levels (using phase 2b PD data only). Tezepelumab PK were well described by a 2-compartment model with first-order absorption; PK parameter estimates were consistent with those of other immunoglobulin G2 antibodies. PK-PD models predicted that subcutaneous dosing at 210 mg every 4 weeks was associated with ~90% of the maximum drug effect of tezepelumab on AER and FeNO; further dose increases were not expected to result in additional, clinically meaningful treatment benefit. No clinically significant covariates of treatment effects on AER and FeNO were identified. Population PK simulations, exposure-response relationships and safety profiles of tezepelumab at doses up to 280 mg every 2 weeks suggested that no dose adjustment based on body weight or for adolescents was required. These results support the selection of 210 mg every 4 weeks subcutaneously as the dose for phase 3 studies of tezepelumab in adults and adolescents with severe asthma.

Keywords

exposure-response, pharmacokinetic-pharmacodynamic modeling, severe asthma, tezepelumab

Thymic stromal lymphopoietin (TSLP) is an epithelial cell-derived cytokine found in the airways that promotes inflammatory responses to airborne particles, such as viruses, allergens, and pollutants, through its activities on multiple pathways.^{1–3} Tezepelumab is a human monoclonal antibody (immunoglobulin [Ig] G2λ) that binds with high affinity to TSLP, preventing it from interacting with its heterodimeric receptor.^{4,5} Tezepelumab is in development for the treatment of multiple inflammatory diseases, including severe asthma, chronic obstructive pulmonary disease, and atopic dermatitis.

Tezepelumab has been studied in multiple phase 1 and 2 trials to evaluate the safety, tolerability, pharmacokinetics (PK), pharmacodynamics (PD), efficacy, and immunogenicity of subcutaneous and intravenous doses in healthy volunteers, patients with severe, atopic dermatitis, adolescents and adults with mild, atopic asthma, and adults with severe asthma.^{4–9} In the phase 2b PATHWAY study, tezepelumab 70 mg every 4 weeks, 210 mg every 4 weeks, and 280 mg every 2 weeks subcutaneously significantly reduced asthma exacerbations by 62%, 71%, and 66%, respectively,

compared with placebo in adults with severe, uncontrolled asthma and improved lung function, asthma control, and patient health-related quality of life.⁴ In addition, reductions from baseline in levels of

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Table 1. Summary of Clinical Studies Included in the Tezepelumab Population PK Model

Study Identifier	Phase	Part	n	Study Population	Doses
NCT00757042 ⁶	1a	A	48	Healthy adults	2.1, 7, 21, 70, 210, and 420 mg SC 210 mg and 700 mg single dose IV
NCT00972179 ⁵	1b	B	9	Adults with atopic dermatitis	700 mg single dose IV
		—	37	Healthy adults	35, 105, and 210 mg Q4W SC 210 mg Q2W SC and 210 mg QW SC
NCT01405963 ⁵	1b	—	16	Adults with atopic asthma	700 mg Q4W IV
NCT01913028 ⁸	I	—	18	Healthy Japanese adult men	Single doses of 35, 105, and 280 mg SC
NCT02512900 ⁷	I	—	21	Adolescents with mild asthma	Single dose of 140 mg SC
NCT02525094 (ALLEVIAD) ⁹	2a	—	51	Adults with atopic dermatitis	280 mg Q2W SC
NCT02054130 (PATHWAY) ⁴	2b	—	393	Adults with severe, uncontrolled asthma	70 mg and 210 mg Q4W SC 280 mg Q2W SC

IV, intravenously; PK, pharmacokinetic; Q2W, every 2 weeks; Q4W, every 4 weeks; QW, every week; SC, subcutaneously.

fractional exhaled nitric oxide (FeNO), eosinophils, and IgE (biomarkers of type 2 inflammation) were observed in all 3 tezepelumab dose groups compared with in the placebo group.⁴ Tezepelumab has been well tolerated at all doses studied thus far (≤ 280 mg every 2 weeks subcutaneously and 700 mg every 4 weeks intravenously), with a low incidence of immunogenicity: posttreatment antidrug antibodies (ADAs) against tezepelumab have been detected in a total of 10 participants across all previous phase 1 and 2 studies.

The objective of this analysis was to use PK-PD modeling to guide tezepelumab dose selection for phase 3 trials in patients with severe asthma.

Methods

All studies were performed in accordance with the ethical standards of the responsible committee on human experimentation and with the principles outlined in the Declaration of Helsinki.

Population PK Modeling

Data from 7 clinical studies were used to develop a population PK model of tezepelumab comprising 5 phase 1 studies that enrolled healthy adults,^{6,8} adults with moderate to severe atopic dermatitis,⁶ adults with mild, atopic asthma⁵, and adolescents with mild to moderate asthma⁷; 1 phase 2a study in adults with moderate to severe atopic dermatitis⁹; and 1 phase 2b study in adults with severe, uncontrolled asthma⁴ (Table 1).

Population PK modeling and simulations were performed using nonlinear mixed-effects modeling software (NONMEM, version 7.3 or higher; ICON Development Solutions, Ellicott City, Maryland).¹⁰ The first-order conditional estimation with the INTERACTION option was employed to maximize the likelihood of model parameters with respect to the observed data for the population PK model.

The base population PK model for tezepelumab consisted of a 2-compartment model with first-order absorption and first-order elimination; a variance component characterizing interindividual variability for clearance (CL), central volume of distribution (V_c), and absorption rate (K_a); and residual unexplained variability, modeled using a proportional and additive model with the additive portion fixed to the lower limit of quantification of the assay (10 ng/mL).

Covariate modeling was conducted to identify the effects of demographics, baseline characteristics, and disease status on tezepelumab PK. Specifically, the covariates evaluated using a stepwise covariate selection approach were age (as a continuous variable), age category (adolescent [12 to 17 years old] vs adult [≥ 18 years old]), sex, body weight, disease status (healthy, atopic dermatitis, or asthma), race (white, Asian, black or African American, or other), and Japanese or non-Japanese ethnicity. Immunogenicity was not tested as a covariate given the low ADA incidence of tezepelumab and the lack of apparent impact of ADAs on PK, based on observed data. A forward selection ($P < .01$) method and a backward elimination ($P < .001$) method were employed to identify statistically significant covariates.

The model was selected based on assessment of goodness-of-fit plots, reduction in objective function for hierarchical models, reduction in interindividual variability, and plausibility of the model parameter value and visual predictive check (VPC). VPCs were used to qualify the ability of the final model to describe the observed data. This was done by simulating 100 replicates of the input data set using the model ($\approx 60\,000$ individuals).

PK-PD Modeling for Asthma Exacerbation Rate and FeNO
PK-PD modeling was performed using the number of asthma exacerbations and FeNO levels over time during the treatment period from the phase 2b

PATHWAY study in patients with severe, uncontrolled asthma. Data from both placebo- ($n = 138$) and tezepelumab-treated ($n = 393$) patients were included in the modeling. Individual tezepelumab concentrations over time, predicted using post hoc PK parameters from the final population PK model, were used as the input to the PK-PD models. The predictive performance of the PK-PD models was assessed using VPCs. In total, 1000 simulated trial replicates were generated using the model parameter estimates and individual covariate values, and dosing regimens from the PATHWAY study.

PK-Asthma Exacerbation Rate Model

A longitudinal, dynamic (nonhomogeneous) Poisson-lognormal count model¹¹ was used to characterize the asthma exacerbation rate (AER) over time. The number of asthma exacerbations in each specified time interval (0-14, 15-28, 29-40, and 41-52 weeks of the phase 2b PATHWAY study) was calculated for each individual and used to support construction of the model.

The log of asthma exacerbation hazard at a given time ($\log[h(t)]$) was expressed as the log of baseline risk of having an asthma exacerbation in the absence of drug (LBase) plus the change in exacerbation hazard because of the drug, expressed as a function of the predicted tezepelumab concentration at time t ($Cp(t)$), which was described using an $E_{\max}$ -type relationship:

$$\log(h[t]) = \text{LBase} + \frac{\log(E_{\max}) \cdot Cp(t)}{EC_{50} + Cp(t)} \quad (1)$$

where $h(t)$ is the exacerbation hazard at time t , LBase is the log-transformed baseline hazard in the absence of drug, $E_{\max}$ is the maximum effect of drug on the hazard of exacerbation (as a ratio over the baseline hazard) at the infinite individual predicted tezepelumab concentration, EC_{50} is the tezepelumab concentration producing half the maximal drug effect, and $Cp(t)$ is the individual predicted tezepelumab concentration at time t . The PK-AER modeling was conducted in NONMEM (version 7.3 or higher)¹⁰ using the Laplace method.

PK-FeNO Model

A population PK-PD model was developed to characterize the relationship between tezepelumab concentration and FeNO measurements. Given that no delay in FeNO response was observed by the time of the first measurement posttreatment (4 weeks), a direct response model was used to describe the relationship between tezepelumab concentration and FeNO level. The FeNO data were log-transformed before model fitting owing to large variability and skewedness in the FeNO data.

A mathematical description of the base FeNO model is provided in the following equation:

$$\log(\text{FeNO}[t]) = \log(\text{base}) + \frac{\log(E_{\max}) \cdot Cp(t)}{EC_{50} + Cp(t)} + \varepsilon \quad (2)$$

where $\text{FeNO}(t)$ is the FeNO level at time t , base is the baseline FeNO level in the absence of drug, $E_{\max}$ is the maximum effect of drug (as a ratio of baseline level), EC_{50} is the tezepelumab concentration producing half the maximal drug effect, $Cp(t)$ represents the individual predicted tezepelumab concentration at time t , and ε is the residual variability. Interindividual variability in baseline FeNO and $E_{\max}$ was also incorporated into the model. The PK-FeNO modeling was conducted in NONMEM (version 7.3 or higher)¹⁰ using the first-order conditional estimation with the INTERACTION option.

Analysis of Covariates in the PK-PD Models

A forward selection ($P < .01$) method and a backward elimination ($P < .001$) method were employed to identify statistically significant covariates for the PK-AER and PK-FeNO models, namely, baseline biomarker levels and disease characteristics that may affect the efficacy of tezepelumab. The covariates evaluated were: number of asthma exacerbations in the previous year (≥ 3 or < 3), oral corticosteroid use (yes or no), inhaled corticosteroid maintenance dose (medium [250-500 μg per day of fluticasone dry powder inhaler or equivalent] or high [> 500 μg per day of fluticasone dry powder inhaler or equivalent]), total IgE at baseline, allergic status (defined as a positive IgE fluorescence enzyme immunoassay test at baseline⁴; yes or no), blood eosinophil count by central laboratory (≥ 250 or < 250 cells/ μL), type 2 helper T (Th2) status (high [serum total IgE > 100 IU/mL and a blood eosinophil count ≥ 140 cells/ μL] or low [patients who did not meet the definition of Th2-high]), forced expiratory volume in 1 second reversibility, FeNO level at baseline, serum periostin level at baseline, and serum dipeptidyl peptidase-4 level at baseline. The potential covariates were tested for impact on baseline exacerbation hazard or FeNO level, as well as on the $E_{\max}$ parameters for both models.

Simulations for Dose Selection

The final PK-PD models were used to predict exposure-response relationships between tezepelumab concentration and AER and FeNO responses. Mean percentage reductions in AER and FeNO across a range of tezepelumab concentrations and the associated 90% confidence intervals (CIs) were simulated using the population mean and the variance-covariance matrix

Table 2. Summary of Covariates for the PK Data Set and for the PK-PD Data Set

Continuous Covariates	PK Data Set (All Studies) Tezepelumab-Treated Participants		PK-PD Data Set (Phase 2b PATHWAY Study) Tezepelumab- or Placebo-Treated Participants	
	n	Mean (SD)	n	Mean (SD)
Age (years)	592	44.9 (15.2)	531	51.6 (12.1)
Baseline body weight (kg)	592	77.7 (16.2)	531	79.0 (15.6)
Height (cm)	592	169 (9.6)	531	167 (9.5)
Baseline FeNO level (ppb)	—	—	527	34.2 (38.2)
Baseline eosinophil count (cells/ μ L)	—	—	531	373 (356)
Baseline FEV ₁ reversibility (mL)	—	—	531	395 (364)
Baseline total IgE (IU/mL)	—	—	531	420 (1100)
Baseline periostin level ng/mL	—	—	526	23.0 (10.9)
Baseline DPP-4 level (ng/mL)	—	—	526	254 (66.1)
Categorical Covariates	Group		Group	
		n (%)		n (%)
Sex	Male	284 (48.0)	Male	184 (34.7)
	Female	308 (52.0)	Female	347 (65.3)
Race	White	477 (80.6)	White	485 (91.3)
	Asian	37 (6.3)	Asian	19 (3.6)
	Black or African American	21 (3.5)	Black or African American	19 (3.6)
	Other	57 (9.6)	Other	8 (1.5)
	Japanese	30 (5.1)	Japanese	19 (3.6)
Japanese ethnicity	Non-Japanese	562 (94.9)	Non-Japanese	512 (96.4)
	Adolescent	21 (3.5)	Adolescent	0 (0.0)
Adolescent vs adult	Adult	571 (96.5)	Adult	531 (100)
	Healthy	102 (17.2)	Healthy	0 (0.0)
Disease status	Atopic dermatitis	60 (10.1)	Atopic dermatitis	0 (0.0)
	Asthma	430 (72.6)	Asthma	531 (100)
OCS use	—	—	No	481 (90.6)
	—	—	Yes	50 (9.42)
Baseline peripheral blood eosinophil count	—	—	≥ 250 cells/ μ L	300 (56.5)
	—	—	< 250 cells/ μ L	231 (43.5)
Number of asthma exacerbations in the previous year	—	—	1 or 2	423 (79.7)
	—	—	≥ 3	108 (20.3)
Maintenance ICS dose ^a	—	—	High	260 (49.0)
	—	—	Medium	271 (51.0)
Baseline Th2 status ^b	—	—	Low	267 (50.8)
	—	—	High	259 (49.2)
Baseline allergic status by FEIA IgE	—	—	Allergic	288 (58.2)
	—	—	Nonallergic	207 (41.8)

DPP-4, dipeptidyl peptidase-4; FEIA, fluorescence enzyme immunoassay; FeNO, fractional exhaled nitric oxide; FEV₁, forced expiratory volume in 1 second; ICS, inhaled corticosteroids; IgE, immunoglobulin E; OCS, oral corticosteroids; PD, pharmacodynamic; PK, pharmacokinetic; ppb, parts per billion; SD, standard deviation; Th2, type 2 helper T.

^a Medium-dose ICS: 250–500 μ g/d of fluticasone dry power inhaler or equivalent; high-dose ICS: > 500 μ g/d of fluticasone dry power inhaler or equivalent.

^b Th2-high: total IgE > 100 IU/mL and a blood eosinophil count ≥ 140 cells/ μ L; Th2-low, patients who did not meet the definition of Th2-high.

of parameter estimates, based on 2000 Monte Carlo simulations.

For the phase 3 dose selection assessment, population PK simulations, performed using the final population PK model, were used to predict the steady-state average tezepelumab concentration ($C_{avg,ss}$) for each of the 3 phase 2b dosage regimens (70 mg every 4 weeks, 210 mg every 4 weeks, and 280 mg every 2 weeks subcutaneously).⁴ The corresponding percentages of maximum FeNO and AER response for

each regimen were predicted based on the respective PK-PD models. The percentages of patients with $C_{avg,ss}$ above the concentrations corresponding to 80% (EC_{80}) and 90% (EC_{90}) of the maximal drug effect based on the PK-AER and PK-FeNO model predictions were calculated. For each of the 3 doses, 100 trials with 200 individuals per trial were simulated. The covariate (including body weight) distributions from the PATHWAY study (Table 2) were used in the simulations.

Table 3. Estimated Tezepelumab PK Parameters From the Final Population PK Model

Parameter	Estimate (% RSE)	95%CI ^a	Interindividual Variability (% RSE) [CV%]	Shrinkage (%)
CL, L/day	0.201 (4.7)	0.182-0.22	0.163 (22.1) [40.4]	2.01
V _c , L	4.28 (6.0)	3.78-4.78	0.191 (21.5) [43.7]	9.55
CLD, L/day	0.726 (9.2)	0.595-0.857	—	—
V _p , L	2.29 (4.5)	2.09-2.47	—	—
K _a , L/day	0.247 (5.3)	0.221-0.273	0.488 (16.1) [69.9]	45.2
Bioavailability	0.645 (4.2)	0.592-0.698	—	—
Effect of body weight on ^b :				
CL	0.957 (8.1)	0.806-1.108	—	—
V _c	0.917 (14.4)	0.658-1.176	—	—
V _p	0.695 (21.6)	0.401-0.989	—	—
Effect of adolescent age on ^c :				
CL	−0.354 (10.2)	−0.425 to −0.283	—	—
V _c	−0.554 (9.6)	−0.659 to −0.449	—	—
Effect of disease status relative to asthma on ^c :				
CL	Healthy: −0.213 (13.6) Atopic dermatitis: −0.101 (40.3)	Healthy: −0.27 to −0.156 Atopic dermatitis: −0.181 to −0.021	—	—
V _c	Healthy: −0.397 (7.7) Atopic dermatitis: −0.164 (27.1)	Healthy: −0.457 to −0.337 Atopic dermatitis: −0.251 to −0.077	—	—
Correlation of individual variability estimate between CL and V _c	0.146 (24.2)	—	—	—
Additive RSE (ng/mL) ^d	10.0 (fixed)	—	—	—
Proportional residual variability (CV%)	19.6 (6.8)	—	—	10.9

BW, body weight; CI, confidence interval; CL, clearance; CLD, distribution clearance; CV, coefficient of variation; K_a, absorption rate constant; LLOQ, lower limit of quantification; PK, pharmacokinetic; RSE, residual error; SE, standard error; V_c, central volume; V_p, peripheral volume.

^aThe 95%CI was calculated using the mean $\pm$ 1.96 $\times$ SE.

^bCovariate effect of body weight was modeled as: $\theta = \theta_{typ} \times (BW/70)^\gamma$, where θ is the PK parameter value and θ_{typ} is the typical value of θ at the reference BW (70 kg), and γ is the power coefficient that represents the covariate effect of BW on θ .

^cFor categorical covariates, the covariate effects were modeled as: $\theta = \theta_{typ} (1 + \theta_k)$, where θ_k represents the covariate effect on the PK parameter of interest, calculated as the relative difference between the PK parameter value in the specified covariate category (θ) and the typical parameter value (θ_{typ}) at the reference covariate value (age category = adult; disease status = asthma).

^dFixed to LLOQ of assay.

Additional population PK simulations were performed using the final model to evaluate fixed dosing versus body weight-based dosing and dosing in adolescents. For the adult simulations, $\approx$ 1000 adults with a body weight distribution based on the PATHWAY study (body weight range, 43-149 kg) were simulated. For the adolescent simulations, a weight distribution was simulated assuming a uniform distribution in age from 12 to 17 years and an equal number of girls and boys in each age group, with body weight randomly sampled from the weight distribution for each age group (10 groups in half-year intervals from 12 to 17 years) and sex based on the US Centers for Disease Control and Prevention growth chart.¹² Approximately 50 adolescents per age group per sex were simulated. Body weight < 40 kg was excluded from the simulation result.

Results

Population PK Modeling

Tezepelumab exhibited linear and time-invariant PK, which were well described by a 2-compartment model with first-order absorption following subcutaneous administration. Baseline body weight, age category (adolescent or adult), and disease status (healthy, atopic dermatitis, or asthma) were identified as statistically significant covariates ($P < .001$). These covariates explained less than 15% of the overall variability in tezepelumab PK, as evidenced by a reduction in individual variability from 49.4% (base model; Table S1) to 40.4% (final model; Table 3) for CL and from 58.5% (base model) to 43.7% (final model) for V_c. The other covariates evaluated, including age (as a continuous covariate), sex, race, and Japanese ethnicity, were not identified as significant covariates for

PK. The final model adequately captured tezepelumab PK across different doses, demonstrated by VPCs (Figure 1A and Figures S1-S6) and goodness-of-fit plots (Figure S7). The PK parameters and their variability values were estimated with adequate precision, and the shrinkage values for CL and V_c were low (Table 3).

PK-PD Modeling

The PK-AER and PK-FeNO models described the phase 2b data reasonably well, demonstrated by VPCs (Figure 1B,C) and goodness-of-fit plots (for the PK-FeNO model; Figure S8). Use of an $E_{\max}$ model resulted in improved fit of the data compared with models without drug effects and models incorporating linear concentration effects, indicating that there is evidence of an exposure-response relationship over the range of concentrations evaluated in the phase 2b study and that this relationship approaches a plateau as the drug concentration increases. A theoretical maximum AER reduction of 70.8% (95%CI, 56.5%-80.4%) was estimated with an EC_{50} of 1.71 $\mu\text{g/mL}$ (95%CI, 0.227-12.8 $\mu\text{g/mL}$); see Table 4. For FeNO, the estimated maximum reduction from baseline because of treatment was 27.8% (95%CI, 23.1%-32.2%) and the estimated EC_{50} was 2.50 $\mu\text{g/mL}$ (95%CI, 1.27-4.93 $\mu\text{g/mL}$).

A history of at least 3 exacerbations in the previous year was identified as a significant covariate for baseline exacerbation rate; however, no significant covariates were identified for the drug effect on AER response. Baseline blood eosinophil count and baseline periostin level were identified as significant covariates for baseline FeNO level (ie, a higher baseline FeNO level was associated with a higher baseline blood eosinophil count and a higher level of periostin). Baseline blood eosinophil count was also a statistically significant covariate for the maximum drug effect on FeNO; the 5th (50 cells/ μL) and 95th (990 cells/ μL) percentiles of baseline blood eosinophil count corresponded to an $E_{\max}$ of 0.524-0.801 (27.4% lower and 10.9% higher than the population mean $E_{\max}$, respectively), suggesting a small (<30%) effect of baseline blood eosinophil count on the maximum FeNO response. None of the other covariates tested had a significant impact on tezepelumab effects on AER and FeNO.

Simulations to Support Dose Selection

Based on the PK-AER model, the simulated $C_{\text{avg,ss}}$ corresponding to the 210-mg every-4-week dose was predicted to achieve 93.2% (90%CI, 71.4%-98.7%) of the maximum predicted AER reduction. Similarly, the simulated $C_{\text{avg,ss}}$ corresponding to the 70-mg every-4-week and 280-mg every-2-week doses were predicted to result in 81.9% (90%CI, 45.4%-96.1%) and $\geq 95\%$ of the maximum predicted AER reduction, respectively (Figure 2A1). Based on the PK-FeNO

model, the simulated $C_{\text{avg,ss}}$ at the 70-mg every-4-week, 210-mg every-4-week, and 280-mg every-2-week doses corresponding to 75.5% (90%CI, 63.6%-98.2%), 90.3% (90%CI, 84.0%-99.4%), and $\geq 95\%$ of the maximum estimated FeNO reduction, respectively (Figure 2B1).

At steady state after doses of 70 mg every 4 weeks, 210 mg every 4 weeks, and 280 mg every 2 weeks, approximately 61%, 100%, and 100% of individuals (Figure 2A2) and 28%, 97%, and 100% of individuals (Figure 2B2) had $C_{\text{avg,ss}}$ greater than or equal to the EC_{80} for AER (6.8 $\mu\text{g/mL}$) and FeNO level (10 $\mu\text{g/mL}$), respectively. The proportions of individuals with $C_{\text{avg,ss}}$ greater than or equal to the EC_{90} were approximately 6%, 83%, and 100% for AER (Figure 2A3) and 1%, 53%, and 99% for FeNO (Figure 2B3) based on the estimated EC_{90} for AER (15.4 $\mu\text{g/mL}$) and FeNO level (22.5 $\mu\text{g/mL}$).

Simulations were also performed with the population PK model to evaluate the need for dose adjustments according to body weight or in adolescent patients. The simulated overall population PK profiles were similar when dosing was fixed (210 mg every 4 weeks) and when dosing was based on body weight (3 mg/kg every 4 weeks; Figure 3A1). Although fixed dosing resulted in increased variation in exposure (area under the curve at steady state) across body weight quartiles compared with body weight-based dosing (Figure 3A2), the simulated overall variability (coefficient of variation [CV]) in exposure was similar for fixed dosing and body weight-based dosing at 46% and 42%, respectively. In addition, the simulated exposure to tezepelumab was lower across all body weight quartiles than that observed for the 280-mg every-2-week dose, which was the highest dose tested and was well tolerated in the phase 2b PATHWAY study with a 52-week treatment duration.⁴

The simulated PK profiles after 210-mg every-4-week dosing in adults versus adolescents indicated that serum concentrations were higher in adolescents (Figure 3B1). At 210 mg every 4 weeks, the mean exposure to tezepelumab in the adolescent population (body weight ≥ 40 kg) was predicted to be 1.41-fold higher than that in adults (based on the effect of both body weight and age category on the CL of tezepelumab observed in the population PK model). However, the simulated exposure to tezepelumab in the adolescent population at 210 mg every 4 weeks was 2.12-fold lower on average (1.71- to 2.65-fold across the body weight quartiles) than that observed in adults at 280 mg every 2 weeks (Figure 3B2).

Discussion

In this study, population PK and PK-PD modeling and simulation were performed to guide tezepelumab dose

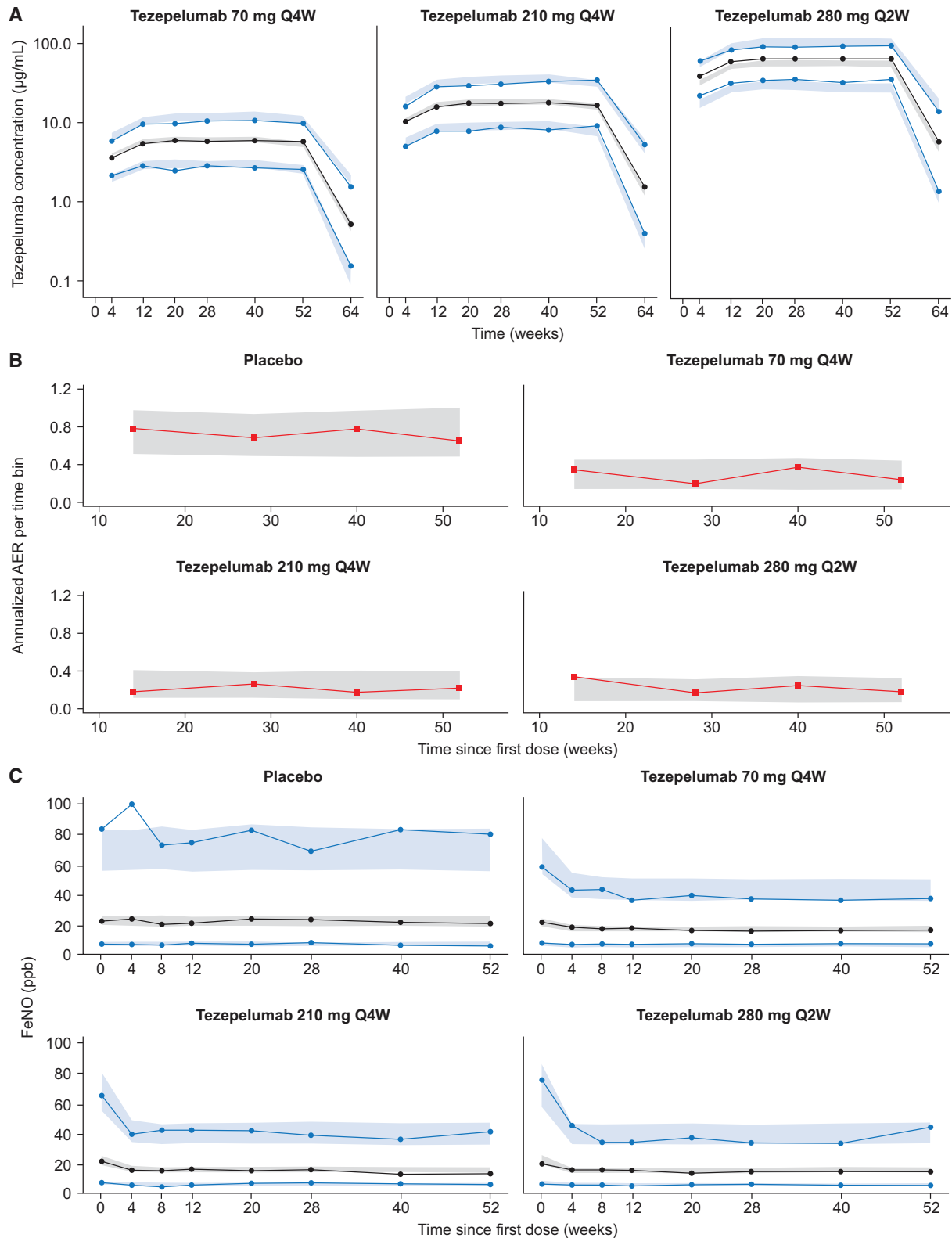


Figure 1. Visual predictive checks of (A) tezepelumab serum concentrations, (B) longitudinal annualized asthma exacerbation rate, and (C) longitudinal FeNO response from the phase 2b PATHWAY study. A and C: Black lines show the observed median and the gray-shaded regions represent the 90% confidence intervals of the model-predicted median responses. Blue lines show the observed 10th and 90th percentiles, and the blue-shaded regions represent the corresponding 90% confidence intervals of the model-predicted 10th and 90th percentiles. B: Solid red squares and red lines show the observed annualized AER, and gray-shaded regions represent the 90% confidence intervals of the model-predicted annualized AER. AER, asthma exacerbation rate; FeNO, fractional exhaled nitric oxide; ppb, parts per billion.

Table 4. Estimated Tezepelumab PK Parameters From the Final PK-AER and PK-FeNO Models

Parameters	Estimate (% RSE)	95%CI ^a	Interindividual Variability (% RSE) [CV%]
PK-AER model			
Baseline hazard (log hazard)	−6.73 (1.87)	−6.98 to −6.48	—
E _{max} (ratio)	0.292 (16.5 ^b)	0.196–0.435	—
EC ₅₀ (μg/mL)	1.71 (193 ^b)	0.23–12.8	—
Effect of ≥3 prior exacerbations on baseline hazard (log hazard) ^c	1.46 (9.93)	1.18–1.744	—
PK-FeNO model			
Baseline FeNO (ppb)	21.1 (1.1 ^b)	19.7–22.6	0.483 (7.24) [69.5]
E _{max} (ratio)	0.722 (9.9 ^b)	0.678–0.769	0.125 (22.6) [35.4]
EC ₅₀ (μg/mL)	2.50 (37.9 ^b)	1.27–4.93	—
Residual variability (CV%)	35.7 (1.30)	34.8–36.6	—
Effect of baseline eosinophil count on:			
Baseline FeNO (log ppb/cell/mL) ^d	0.00076 (13.2)	5.64×10^{-4} to 9.56×10^{-4}	—
E _{max} (1/cell/mL) ^e	−0.00045 (16.9)	$-6.03E \times 10^{-4}$ to -3.03×10^{-4}	—
Effect of baseline periostin on baseline FeNO (log ppb/ng/mL) ^d	0.010 (26.2)	4.90×10^{-3} to 1.51×10^{-2}	—
Correlation of individual variability estimate between baseline and E _{max}	−0.168 (16.0)	−0.221 to −0.115	—

AER, asthma exacerbation rate; CI, confidence interval; CV, coefficient of variation; EC₅₀, drug concentration producing half of the maximal drug effect; E_{max}, maximal drug effect; FeNO, fractional exhaled nitric oxide; PK, pharmacokinetic; ppb, parts per billion; RSE, residual error; SE, standard error.

^aThe 95%CI was calculated using the mean $\pm$ 1.96 $\times$ SE.

^b% RSE was based on the log-transformed parameter value.

^cThe covariate effect of ≥ 3 prior exacerbations on baseline hazard was modeled as: $LBASE = LBASE_{typ} + \theta_k$, where $LBASE_{typ}$ is the typical baseline hazard value of an individual with <3 prior exacerbations, and θ_k is the difference in baseline hazard value for an individual with ≥ 3 prior exacerbations compared with the typical value.

^dThe covariate effect of a given biomarker on baseline FeNO level was modeled as $\log(BASE) = \log(BASE_{typ}) + \theta_k \cdot (cov - cov_{ref})$, where BASE is the baseline FeNO value, $BASE_{typ}$ is the typical value of FeNO baseline at the reference covariate, and θ_k is the coefficient that represents the covariate effect on the BASE.

^eThe covariate effect of a given biomarker on the maximum effect was modeled as: $\log(E_{max}) = \log(E_{max,typ}) + \theta_k \cdot (cov - cov_{ref})$, where E_{max} is the maximum effect value, $E_{max,typ}$ is the typical value of the maximum effect at the reference covariate, and θ_k is the coefficient that represents the covariate effect on the E_{max} .

selection for phase 3 trials in adults and adolescents with severe asthma. Tezepelumab PK was well described by a 2-compartment model with first-order absorption. No target-mediated drug disposition was observed, and the parameter estimates obtained were similar to values for other IgG2 monoclonal antibodies that do not exhibit target-mediated disposition.¹³ Baseline body weight, age, and disease status were significant covariates of exposure to tezepelumab; however, these covariates only explained a small proportion (<15%) of the overall variability in tezepelumab PK.

An effect of body weight on PK is commonly observed for monoclonal antibodies.¹⁴ The estimated body weight exponent for CL of tezepelumab from the final model (0.957) is within the range that has been previously reported for other monoclonal antibodies.¹⁴ The model-estimated CL and V_c were numerically lower in adolescents than in adults; however, it must be noted that estimates of higher tezepelumab exposure in adolescents could be confounded by between-trial differences, because PK data for adolescents were obtained from a separate trial with a small number of adolescent participants ($n = 21$). The difference in tezepelumab PK between adolescents and adults

requires further investigation when more adolescent data are available from the phase 3 studies.

Disease status (healthy, atopic dermatitis, or asthma) was also identified as a statistically significant covariate for CL of tezepelumab. Higher CL of tezepelumab was observed in patients with atopic dermatitis and asthma than in healthy individuals, which may be attributed to a potentially higher protein catabolic rate associated with inflammation. However, the overall impact on exposure to tezepelumab is not considered clinically meaningful because of a small magnitude of effect (10.1%–21.3%) compared with the overall variability in CL (CV% of 40.4%). In addition, the proportions of healthy individuals (17.2%) and individuals with atopic dermatitis (10.1%) were much smaller than the proportion of participants with asthma (72.7%), which may affect the accuracy of the estimated effect of disease status on tezepelumab PK. This may be further confounded by intertrial variability. The effect of disease status on tezepelumab PK needs to be evaluated further when more PK data are available from healthy volunteers and patients with atopic dermatitis.

It should be noted that an exhaustive evaluation of intrinsic and extrinsic factors that may affect PK was

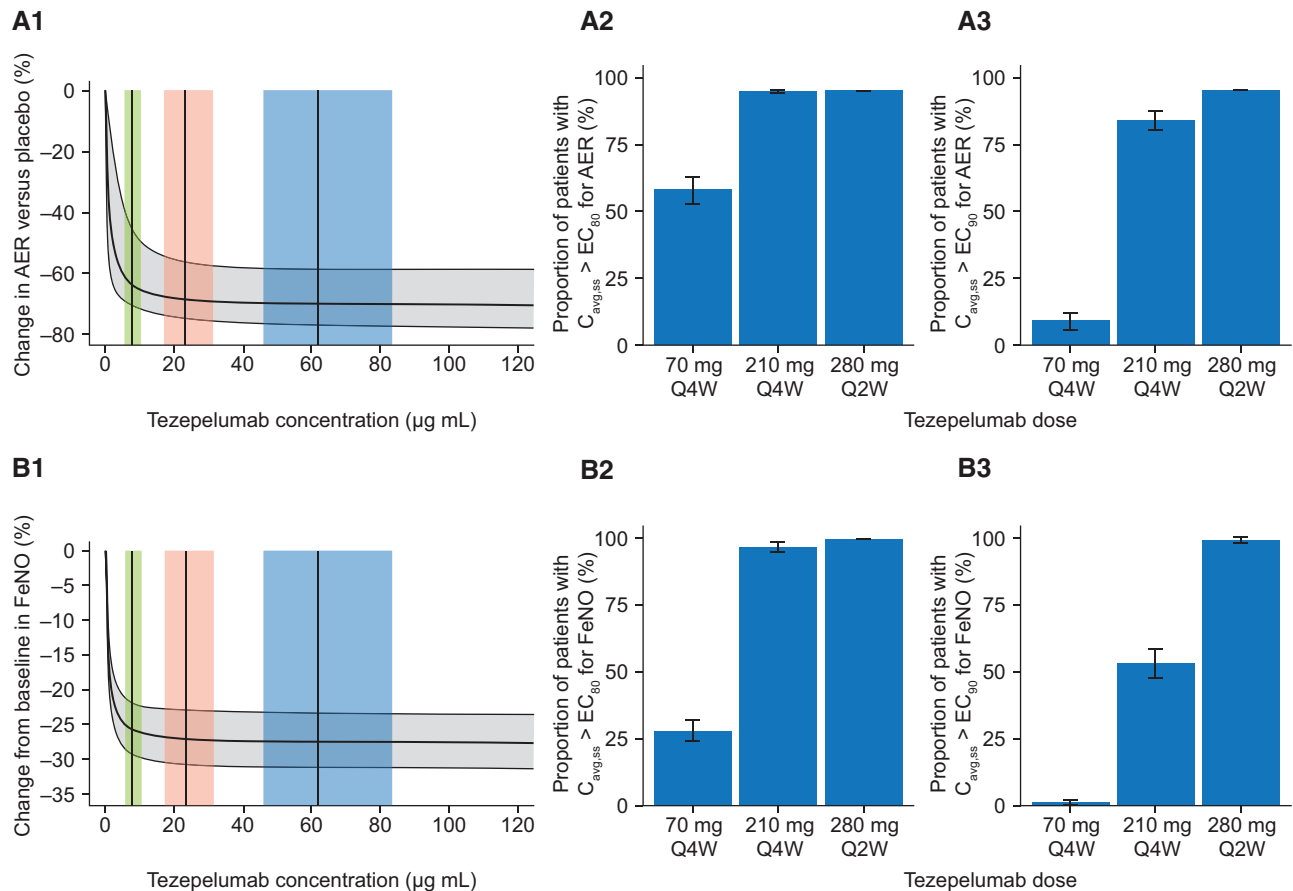


Figure 2. Exposure-response relationship for tezepelumab and the predicted proportions of patients achieving steady-state average concentrations greater than the EC_{80} and EC_{90} . (A1, B1) Tezepelumab concentration-response relationships for AER and FeNO reduction, respectively. The solid black line and gray shading show the simulated mean percentage reductions in AER and FeNO, respectively, across a range of tezepelumab concentrations, and the associated 90% confidence intervals reflect the uncertainties in the PK-PD model parameters. The green, red, and blue boxes show the model-predicted medians and interquartile ranges of simulated $C_{avg,ss}$ for tezepelumab 70 mg every 4 weeks, 210 mg every 4 weeks, and 280 mg every 2 weeks, respectively, from the PK model (100 trial simulations with 200 individuals per trial). (A2, A3) Proportions of individuals with $C_{avg,ss} > EC_{80}$ (6.84 μg/mL) and $> EC_{90}$ (15.4 μg/mL), respectively, for AER. (B2, B3) Proportions of individuals with $C_{avg,ss} > EC_{80}$ (10 μg/mL) and $> EC_{90}$ (22.5 μg/mL), respectively, for FeNO. AER, asthma exacerbation rate; $C_{avg,ss}$, average concentration at steady state; EC_{80} , drug concentration producing 80% of the maximal drug effect; EC_{90} , drug concentration producing 90% of the maximal drug effect; FeNO, fractional exhaled nitric oxide; PK, pharmacokinetic.

not performed for this analysis because the primary goal was to provide individual PK predictions for PK-PD modeling to enable dose simulations (including adolescent and fixed versus body weight-based dosing) to support phase 3 dose selection. The population PK model will be updated with data from the phase 3 program, when additional covariates will be evaluated based on a larger data set. Nevertheless, this is not expected to impact the modeling and simulation results or conclusions from this analysis significantly.

The PK-AER and PK-FeNO models described the AER and FeNO data from the phase 2b PATHWAY study well, across all doses. The model parameters were generally estimated with good precision, except for the EC_{50} value of AER. This was expected because of the sparse exacerbation data and is unlikely to affect

the overall conclusion from the PK-PD modeling and simulations. PK-PD modeling and simulations indicated that the average concentrations of tezepelumab at steady state that corresponded to 70 mg every 4 weeks, 210 mg every 4 weeks, and 280 mg every 2 weeks were associated with 75.5% to 81.9%, 90.3% to 93.2%, and $>95\%$, respectively, of the maximum drug effect on AER and FeNO. These results suggest that the 210 mg every-4-week dose is at the plateau of the exposure-response curve for tezepelumab and that a further increase in dose is not expected to result in additional clinically meaningful treatment benefit. The 70-mg every-4-week dose is close to but not entirely on the plateau of the exposure-response curve and therefore is expected to render a suboptimal effect on AER and FeNO. At the dose of 210 mg every

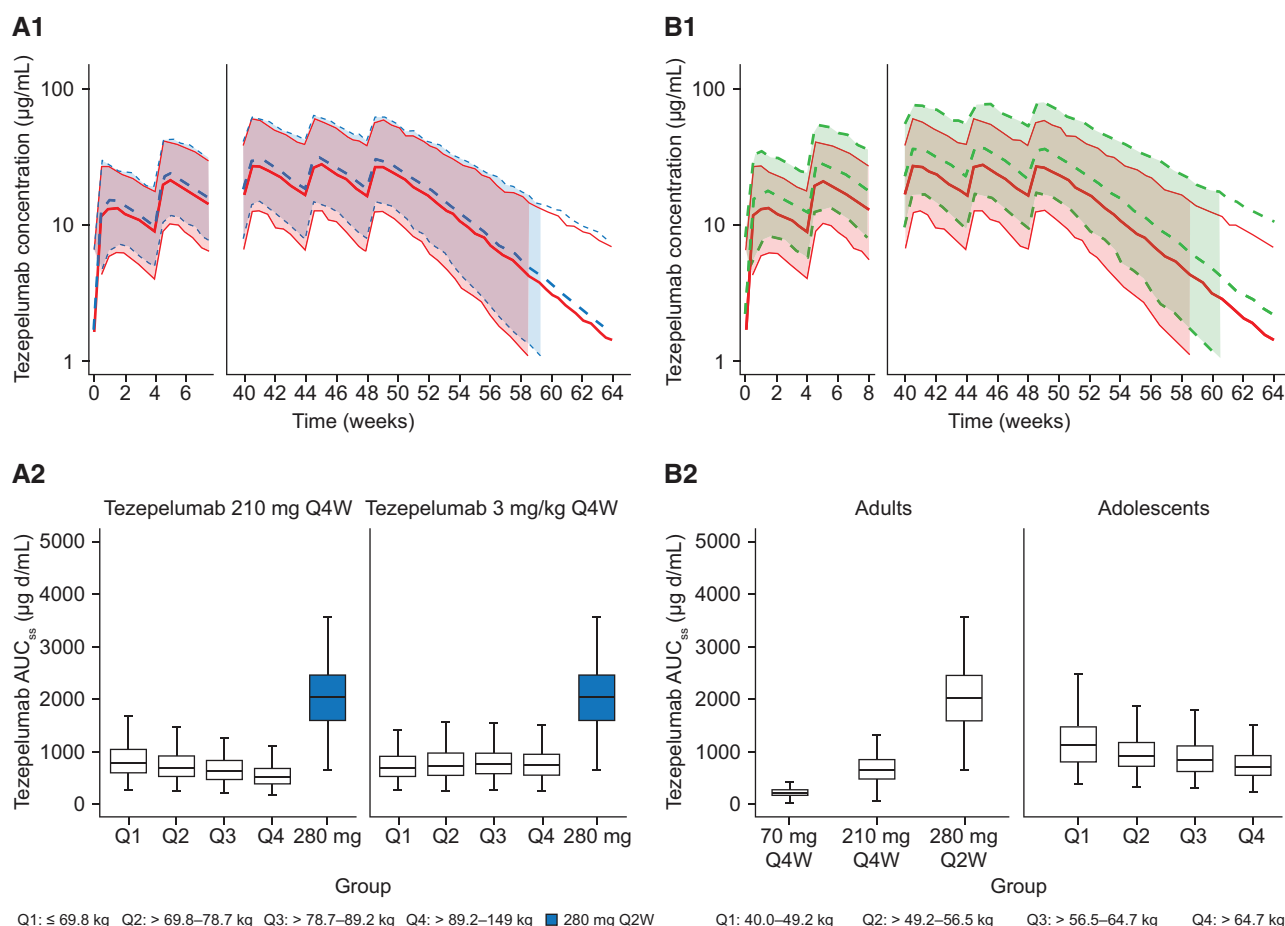


Figure 3. PK simulations for tezepelumab. (A1) Simulated concentration-time profiles after fixed and body weight-based dosing. The model-predicted median (5th–95th percentiles) serum tezepelumab concentration over time after 210 mg (shaded red) versus 3 mg/kg (shaded blue) every 4 weeks from ≈ 1000 simulated profiles is shown. The 5th, 50th, and 95th percentiles for 210 mg and 3 mg/kg every 4 weeks are shown by the solid red lines and blue dashed lines, respectively. (A2) Box-and-whiskers plots showing the predicted median (horizontal line), 25th–75th quartile range (box), and minimum and maximum (whiskers) distribution of exposure (AUC_{ss}) across weight quartiles for fixed and body weight-based dosing ($n = 1000$ simulated individuals). The model-interpolated AUC_{ss} distribution at the highest dose of 280 mg every 2 weeks in the phase 2b study is also shown for comparison. (B1) Simulated concentration-time profiles in adults and adolescents. The model-predicted median (5th–95th percentiles) serum tezepelumab concentration after 210 mg in adults (shaded red) versus 210 mg in adolescents (shaded green) from ≈ 1000 simulated profiles is shown. The 5th, 50th, and 95th percentiles for adults and adolescents are shown by the solid red lines and the green dashed lines, respectively. (B2) Box-and-whiskers plots showing the predicted median (horizontal line), 25th–75th percentile range (box), and minimum and maximum (whiskers) distribution of exposure (AUC_{ss}) in adolescents across body weight quartiles from ≈ 1000 simulated individuals. The model-interpolated AUC_{ss} at doses of 70 mg every 4 weeks, 210 mg every 4 weeks, and 280 mg every 2 weeks for adults in the phase 2b study are shown for comparison. AUC_{ss} , area under the curve at steady state; PK, pharmacokinetic; Q, quartile.

4 weeks, $>95\%$ of patients were predicted to reach the EC_{80} and $>80\%$ and 50% of patients were predicted to reach the EC_{90} for AER and FeNO, respectively. By contrast, only 61% and 28% of patients were predicted to achieve EC_{80} for AER and FeNO, respectively, at the 70-mg every-4-week dose, suggesting that 70 mg every 4 weeks is unlikely to provide optimal efficacy in all individuals. Although the proportions of participants achieving the EC_{90} was greater for the 280-mg every-2-week dose regimen than for the 210-mg every-4-week dose regimen, this was not expected to confer clinically meaningful improvement in efficacy because treatment with the 210-mg every-4-week dose regimen

was predicted to result in $>90\%$ of the maximum drug effect on AER. The model predictions were consistent with the observations from the phase 2b PATHWAY study, which showed that the AER reduction versus placebo was lower in the 70-mg every-4-week group (62%) than in the tezepelumab 210-mg every-4-week group (71%) and did not increase in the 280-mg every-2-week group (66%).⁴ The slightly lower reduction in AER in the 280-mg every-2-week dose group compared with the 210-mg every-4-week dose group was likely because of interindividual variability. The observed data from the PATHWAY study also showed that the reduction from baseline in FeNO was similar across the

3 dose groups,⁴ confirming that all 3 doses were at or close to the plateau region of the exposure-response curve. The suboptimal response at 70 mg every 4 weeks was not observed for FeNO, which was likely because of the small predicted difference in response relative to the interindividual variability as well as any imbalances in patient characteristics between the dose groups (eg, mean baseline FeNO was highest in the 70-mg every-4-week group⁴). Furthermore, safety data from the PATHWAY study showed that tezepelumab was well tolerated with all 3 dose regimens, with a similar incidence of adverse events across treatment groups,⁴ suggesting that exposure had no impact on the safety of tezepelumab. Together, these results support the selection of the 210-mg every-4-week dose for the phase 3 studies of tezepelumab in patients with severe asthma.

In the PK-FeNO model, baseline blood eosinophil count was identified as a statistically significant covariate for the $E_{\max}$ for FeNO; however, the effect on the $E_{\max}$ was <30% for the 5th and 95th percentiles of baseline blood eosinophil counts and was not considered clinically significant. None of the other covariates tested, including protocol-specified biomarkers, had a clinically significant impact on tezepelumab effect, either on AER or FeNO, confirming the broad efficacy of tezepelumab in patients with severe asthma, irrespective of baseline inflammatory biomarker status.^{4,15}

Body weight-based and fixed doses of tezepelumab were compared using population PK simulations based on the body weight distribution of patients from the phase 2b PATHWAY study (range, 43–149 kg). The median simulated exposure to tezepelumab and the overall variability in exposure were similar with body weight-based and fixed dosing, which suggests that both dosing regimens will provide similar efficacy and safety outcomes. Owing to the effect of body weight on tezepelumab PK, body weight-based dosing provided more consistent exposure to tezepelumab across body weight quartiles, whereas fixed dosing resulted in slightly higher exposure to tezepelumab in individuals with a lower body weight and lower exposure to tezepelumab in those with higher body weight. This observation is consistent with that for other compounds for which body weight is a significant covariate for CL.¹⁴ Despite the effect of body weight on exposure to tezepelumab, the simulated exposure across the body weight quartiles at the proposed dose of 210 mg every 4 weeks was well under the exposure at 280 mg every 2 weeks, which was shown to be well tolerated in the phase 2b study.⁴ In addition, because the dose of 210 mg every 4 weeks is at the plateau of the exposure-response curves, the small variation in exposure across the body weight quartiles is not predicted to result in any clinically meaningful difference in efficacy.

Given these findings and in view of the convenience of fixed dosing, a fixed dose of 210 mg every 4 weeks is recommended for the phase 3 studies of tezepelumab.

At a fixed dose of 210 mg every 4 weeks, adolescents were predicted to have a 1.4-fold higher mean exposure to tezepelumab than adults given the same dose, owing to the effect of both body weight and age category on the CL of tezepelumab. Despite this, the mean exposure of the adolescent population (body weight ≥ 40 kg) to tezepelumab after 210-mg every-4-week dosing across body weight quartiles was lower than the exposure of adults to tezepelumab after 280-mg every-2-week dosing. Given that tezepelumab has been shown to be well tolerated at 280 mg every 2 weeks in adults,⁴ the exposure of adolescents to tezepelumab 210 mg every 4 weeks is considered unlikely to pose a safety risk. Considering the convenience of fixed dosing, the dose of 210 mg every 4 weeks subcutaneously was selected for both adults and adolescents with a body weight of 40 kg or higher in the phase 3 studies of tezepelumab.

Conclusion

In conclusion, a population PK model was developed to describe tezepelumab PK in healthy volunteers and patients with asthma or atopic dermatitis and PK-PD models to describe AER and FeNO responses in patients with severe, uncontrolled asthma. The modeling and simulation results supported selection of a fixed-dose regimen of 210 mg every 4 weeks subcutaneously for phase 3 studies of tezepelumab in adults and adolescents with severe asthma, which was subsequently used in the NAVIGATOR (ClinicalTrials.gov identifier: NCT03347279) and SOURCE (ClinicalTrials.gov identifier: NCT03406078) studies.

Conflicts of Interest

Y.Z. and J.M.G. are employees of AstraZeneca. N.L. was an employee of AstraZeneca at the time of the study and is a current employee of NGM Biopharmaceuticals. R.v.d.M. was an employee of AstraZeneca at the time of the study and is a current employee of DBV Technologies. B.A. was an employee of AstraZeneca at the time of the study and is a current employee of Gilead. J.R.P. is an employee of Amgen. L.R. was an employee of AstraZeneca at the time of the study and is a current employee of Exelixis. Medical writing support was provided by Laura Knapp, PhD, of PharmaGenesis London, London, UK, with funding from AstraZeneca and Amgen, Inc.

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Data Availability Statement

Data underlying the findings described in this article may be obtained in accordance with AstraZeneca's

data-sharing policy described at: <https://astrazeneca.grouptrials.pharmacm.com/ST/Submission/Disclosure>

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Supplemental Information

Additional supplemental information can be found by clicking the Supplements link in the PDF toolbar or the Supplemental Information section at the end of web-based version of this article.